

# Mohamed Ragab

AI & Digital Construction Engineer · Structural engineering, BIM & computer vision

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Portfolio: mohamedragab4554.github.io

## PROFILE

I am a structural engineer who builds AI: I engineer the data, train computer-vision and point-cloud models on site photos, structural drawings and laser scans, and deploy them through cloud pipelines into BIM, CAD-to-BIM, Scan-to-BIM and digital-twin workflows. I hold an MSc (Distinction) in Digital Construction Analytics & BIM and have a year of structural and façade design delivery. I am co-founder and CTO of AECAI, an AI-assisted structural inspection platform.

## EXPERIENCE

R&D AI-Construction Specialist · AGECS (remote) Apr 2026 – Present

- Column detection on plans with YOLOv8m-seg: mAP50 0.941, precision 0.944, recall 0.935 (tiled inference, 4-rotation TTA, pixel-evidence false-positive filters).
- 8-class structural-element segmentation (beam, column, opening, pile, wall, ...) on ~21k tiles: validation mAP50 0.902 box / 0.893 mask.
- Human-in-the-loop Scan-to-BIM: a ~264 M-point scan to a verified IFC (2 slabs, 9 walls, 21 columns), with automatic rejection of temporary shoring.

Co-Founder & CTO · AECAI Ltd, Belfast Dec 2025 – Present

- Built the complete Next.js web console: dashboards, assets, inspection forms and builder, photo review, reports.
- Designed the spalling and exposed-rebar U-Net pipeline (patching, Gaussian stitching, TTA) with parallel serverless GPU inference (RunPod, Supabase).
- Corrected a misleading training metric (F1 0.14 → true 0.901, IoU 0.820 for rebar) and ran a false-positive study across 23 test images.

Digital Design Engineer: Industry Collaboration · AECOM × Ulster University Feb 2025 – May 2025

- Revit model of a concrete water tank from AECOM drawings; crack classifier at 89.69% accuracy (VGG16); width classes against EN 1992-1-1.
- Power BI dashboard with an embedded 3D model, crack register and maintenance actions. Module mark 80%.

Structural & Façade Design Engineer (Graduate) · National Consulting Engineers Inc., Florida (remote) Sep 2023 – Sep 2024

- 115+ design packages (glass railings, curtain walls, canopies, stairs, connections in steel, aluminium and timber) to AISC, ASCE 7, ACI and FBC, using ETABS, SAP2000, SAFE, IDEA StatiCa and AutoCAD.

Internships · DME Holding (planning, Primavera P6) · Redcon Construction (site supervision, piles) 2021 – 2023

## RESEARCH

MSc dissertation (2025): Comparative YOLO and U-Net deep learning for multi-defect detection. YOLOv8x-seg box mAP50 0.807; U-Net ResNet-50 mIoU 0.630 / F1 0.775; field recall 0.933 (YOLO11x-seg) on manually verified industry images; 18-class DACL10k segmentation; survey of 21 practitioners. Journal manuscript in submission.

## EDUCATION

MSc Digital Construction Analytics & BIM: Distinction · Ulster University 2024 – 2025

BSc Civil Engineering · Higher Technological Institute, Egypt · GPA 3.24/4.00 · Graduation project A+ 2018 – 2023

## SKILLS

|                      |                                                                                                                                           |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Computer vision & ML | PyTorch, Ultralytics YOLO, U-Net/FPN (SMP), TensorFlow/Keras, PointNet++; evaluation, field validation & error analysis                   |
| Data processing      | Dataset engineering (COCO/YOLO/SVG converters), point clouds (Open3D, ~264 M points), pandas, NumPy, OpenCV, CVAT annotation QA, SQL      |
| Cloud & MLOps        | RunPod serverless GPU, Supabase (Postgres, storage, edge functions), Hugging Face, Docker, Vercel, GitHub Actions CI/CD                   |
| BIM & digital twins  | Revit, Navisworks, AutoCAD, Dynamo, pyRevit, IFC/IfcOpenShell, ReCap, Power BI twins; ISO 19650, BIM Level 2 delivery, Level 3 principles |
| Generative AI        | Multimodal LLM/VLM evaluation on drawings, LLM features in products (Claude, Gemini), coding agents (Claude Code, Codex, Kimi)            |
| Structural           | ETABS, SAP2000, SAFE, IDEA StatiCa, RAM Connection, Tekla; AISC, ASCE 7, ACI, Eurocodes                                                   |

## CERTIFICATIONS & MEMBERSHIPS

ICE Student Member · Deep Learning for Computer Vision (ITI, 2026) · AI Fluency (Anthropic, 2026) · Steel Design Diploma and Concrete Technical Design (Autodesk ATC, 2022) · AUC project-management courses (2023) · Tekla shop drawings (2024)

Languages: Arabic (native), English (fluent)